

EAST WEST UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Program: Information System Analysis and Design CSE-347

Feasibility Study Report

SmartPC Advisor

A rule-based web system for compatible, value-conscious PC build planning

Course Instructor : Dr. Hosney Jahan

Assistant Professor

Prepared by : Group-07

1.	Shimul Kumar	2023-3-60-157
2.	Shakkar Chowdhury	2023-3-60-342
3.	Md. Rakibuzzaman	2023-3-60-615

Submission date: 8 October 2026

Executive Summary

SmartPC Advisor is a proposed web application designed to help users build desktop PCs according to their budget, intended use, performance requirements, and preferences. The system will recommend suitable components, check compatibility using predefined rules, estimate power requirements, and compare prices and performance to help users make informed purchasing decisions.

The project will be developed by a team of three students as part of a three-month university semester project. To ensure successful completion within the available time, the team will focus on a practical prototype with a limited component catalog and essential features. The application will use React or Next.js for the frontend, FastAPI for the backend, and a relational database to store component specifications and prices. The initial version will use rule-based recommendations rather than artificial intelligence (AI) or machine learning (ML), keeping development manageable for the team.

The project is technically feasible because the required technologies and development tools are available. It is also economically feasible for an academic project because the team can use existing computers and free development tools. However, completing the project within three months will require clear task distribution, regular progress reviews, and a well-defined scope. The team will prioritize core features such as component selection, compatibility checking, estimated power requirements, and price-performance comparisons rather than advanced features that could delay completion.

The main challenge will be maintaining reliable component information. The team will collect specifications from trusted sources, define compatibility rules, and record price sources and update dates. Since component availability and prices may change, the system will present prices as dated estimates rather than live market data. Compatibility results will also be treated as guidance, with uncertain cases flagged for manual verification.

Recommendation: GO. SmartPC Advisor is a suitable project for a three-member team working within a three-month semester. The team should begin with a clearly defined feature set, divide responsibilities among the members, and review progress regularly. By focusing on essential features and avoiding unnecessary complexity, the team can aim to deliver a functional prototype that demonstrates the practical value of rule-based PC component recommendations within the semester.

Report Basis and Interpretation

This report evaluates a proposed system, not a completed implementation. No user survey, performance benchmark, vendor quotation, prototype test, or empirical demand study is claimed. All budget, staffing, schedule, and scoring values are planning assumptions for decision-making and should be replaced with institution-specific evidence before project approval.

Contents

Executive Summary.....	2
Report Basis and Interpretation.....	2
Contents.....	3
1. Introduction and Background.....	4
2. Problem Statement and Objectives.....	4
2.1 Problem Statement.....	4
2.2 Main Objective.....	4
2.3 Specific Objectives.....	4
3. Proposed System and Scope.....	5
3.1 User Journey and Major Functions.....	5
3.2 Core Scope for the Initial Release.....	5
3.3 Exclusions and Future Scope.....	6
3.4 Indicative Recommendation Logic.....	6
4. Assumptions and Alternatives.....	7
5. Feasibility Assessment.....	7
5.1 Technical Feasibility.....	7
5.2 Operational Feasibility.....	8
5.3 Economic and Financial Feasibility.....	9
5.4 Schedule Feasibility.....	10
5.5 Legal, Ethical and Data Feasibility.....	11
5.6 Security and Privacy Considerations.....	11
6. Schedule and Resource Feasibility.....	12
6.1 People and Roles.....	12
6.2 Infrastructure and Tools.....	12
7. Legal, Ethical, Data and Security Considerations.....	12
7.1 Data Governance.....	12
7.2 Ethical Use and Accessibility.....	13
8. Risk Assessment.....	13
9. Constraints, Benefits and Limitations.....	14
9.1 Constraints.....	14
9.2 Expected Benefits.....	14
9.3 Limitations.....	14
10. Conclusion and Recommendation.....	15

1. Introduction and Background

Selecting the right parts for a desktop PC can be difficult because buyers must consider their budget, intended use, component compatibility, availability, power requirements, and performance. Even if individual components look good, they may not work together properly. Beginners may also struggle to compare different components and decide which combination offers the best value for their needs.

SmartPC Advisor is a proposed web application that helps users choose suitable PC components based on their budget, needs, and preferences. It will recommend complete PC builds, check component compatibility, estimate power requirements, and compare prices and performance. The first version will use predefined rules and a database of component information instead of artificial intelligence (AI) or machine learning (ML). This approach makes the system easier to develop, test, and explain within our three-month semester project.

This feasibility study evaluates whether a three-member student team can develop the application within the available time and resources. It considers technical, operational, economic, schedule, legal and data, security, and resource feasibility. The study provides an initial assessment of the project's practicality, but it does not guarantee successful completion or prove that there is sufficient market demand for the application.

2. Problem Statement and Objectives

2.1 Problem Statement

A user may know their budget and what they want to do with the computer, such as gaming, programming, or video editing, but may not know which processor, graphics card, RAM, motherboard, storage, power supply, and case to choose. Manual research can be time-consuming, product data may be inconsistent, and compatibility checks can be missed.

SmartPC Advisor aims to solve this problem by helping users select suitable PC components and create a complete build based on their budget and requirements.

2.2 Main Objective

To design and develop a web-based, rule-based system that recommends compatible and cost-conscious PC builds based on budget, intended use, performance requirements, component value, and available product data.

2.3 Specific Objectives

1. **Build a component database:** Store information about processors, graphics cards, RAM, motherboards, storage, power supplies, and cases.
2. **Recommend PC builds:** Suggest suitable combinations of components based on the user's budget and intended use.
3. **Check compatibility:** Identify known compatibility issues between selected components and explain potential problems.
4. **Estimate power requirements:** Calculate the estimated power needs of a build and recommend a suitable power supply with an appropriate safety margin.
5. **Compare price and performance:** Help users compare different builds and understand which offers better value for their needs.

6. **Provide a user-friendly interface:** Allow users to search, filter, view, and compare components and complete PC builds.
7. **Show reliable information:** Display component data sources, last-updated dates, and system limitations so users can make informed decisions.

3. Proposed System and Scope

3.1 User Journey and Major Functions

1. **Enter Requirements:** The user selects a budget, chooses a purpose such as gaming, programming, or general use, and specifies performance needs and component preferences.
2. **Select Components:** The system searches its component database and selects suitable products that match the user's requirements and budget.
3. **Check Compatibility:** The system checks whether the selected components can work together. If a compatibility issue is found or some information is missing, it displays a warning.
4. **Rank Recommended Builds:** The system compares compatible builds based on price, estimated performance, and value for money. It also explains why each build is recommended.
5. **Compare and Customize:** The user can view component details, check estimated power requirements, compare different builds, and change their preferences to explore other options.

3.2 Core Scope for the Initial Release

Area	Features
Component Database	Store specifications and prices for seven types of PC components: CPU, GPU, RAM, motherboard, storage, power supply, and case. Include basic tools for adding and updating component information.
PC Recommendations	Allow users to enter their budget and intended use. Recommend complete PC builds using predefined rules.
Compatibility Checking	Check processor and motherboard compatibility, RAM type and capacity, case size and clearance, power supply wattage and connectors, and storage interfaces. Show warnings when problems or unknowns are found.
Price and Performance Comparison	Display the total build cost, estimated performance, value for money, and alternative builds so users can compare their options.
User Interface	Provide component search, filters, PC build visualization, compatibility warnings, estimated power consumption, and power supply recommendations through a responsive web interface.
Testing and Documentation	Include sample component data, test cases, documented compatibility rules, a user guide, and component data sources with last-updated dates.

3.3 Exclusions and Future Scope

The first version will not include live retailer price updates, guaranteed product availability, exact gaming FPS predictions, automatic bottleneck detection, AI-based recommendations, regional price comparisons, user accounts, saved PC builds, community-submitted components, or a mobile application.

These features may be considered for future development after the core system has been completed and tested. If we later introduce live price updates from third-party sources, we will need to follow their access rules, give proper credit to data sources, and maintain a process for updating and correcting price information.

Our priority is to deliver a working, reliable, and easy-to-use PC recommendation system within the available time. Additional features will be considered only if the core system is completed successfully and sufficient time and resources remain.

3.4 Indicative Recommendation Logic

SmartPC Advisor will recommend PC builds based on the user's budget, intended use, and component compatibility. First, the system will remove or flag builds that exceed the budget or fail essential compatibility checks. The remaining builds will be ranked according to their suitability and value.

The system may use the following scoring model:

Factor	Weight	Purpose
Workload suitability	45%	Measures how well the build matches the user's intended use, such as gaming, programming, or general productivity.
Value for money	30%	Compares estimated performance with the total build cost.
Budget utilization	15%	Evaluates how effectively the build uses the user's available budget without unnecessary spending.
Upgrade potential	10%	Consider opportunities to upgrade components in the future.
Total	100%	

These weights are initial design choices and will be adjusted if testing shows that they produce poor recommendations. The system will explain why each build is recommended and display its component specifications, prices, and total cost.

Compatibility will always come first. A build that fails an essential compatibility check cannot become a valid recommendation simply because it receives a high score.

SmartPC Advisor will estimate a PC's power requirements using available component power specifications or documented estimates. It will then apply a defined safety margin to recommend a suitable power supply unit (PSU). The estimated power requirement will be displayed clearly so users understand that it is an approximation, not a direct measurement of actual electricity consumption. PSU recommendations will also consider other important factors, including connector compatibility, power quality, transient power demands, and appropriate safety certifications.

4. Assumptions and Alternatives

The development of SmartPC Advisor is based on the following assumptions:

1. **Project Duration:** The project will be developed by a team with three members over approximately 12 weeks as part of a university semester course.
2. **Available Resources:** Team members will use their existing computers, internet access, and free development tools to reduce project costs.
3. **Component Database:** The initial database will contain approximately 150–300 component records. The team will collect specifications from public manufacturer sources and review the information before adding it to the database.
4. **Component Prices:** Prices will be recorded in Bangladeshi Taka (BDT), along with their sources and collection dates where possible. The system will not provide guaranteed live prices or complete market coverage.
5. **Project Scope:** The application will focus on desktop PC builds. Laptop recommendations, servers, certified professional workstations, and custom liquid-cooling configurations will be excluded from the first version.
6. **Project Supervision:** The supervisor or project owner will review the project scope, compatibility rules, and data sources at agreed stages.
7. **Cost Estimates:** Development and hosting costs will be estimated using planning figures dated 8 October 2026. Actual costs will be confirmed before any purchases are made.

These assumptions will help the team keep the project realistic, affordable, and achievable within the three-month semester. The catalog size and feature scope may be adjusted according to the team's progress and available time.

5. Feasibility Assessment

5.1 Technical Feasibility

The proposed stack is technically suitable for a moderate-size academic prototype. The frontend can be developed using React or Next.js to provide an interactive user interface. FastAPI can handle the backend logic, including component selection, compatibility checking, and build recommendations. PostgreSQL or MySQL can store component specifications, prices, and compatibility rules. Recharts or Chart.js can display budget distribution and compare different PC builds.

The selection logic can initially run in the application service and be separated from the interface for review and testing.

Technical Area	Feasibility	Explanation
Component Database	High	The team can store component specifications, prices, data sources, and compatibility information in a relational database.

Compatibility Checking	Moderate to High	Predefined rules can identify known compatibility issues. Missing information must be flagged rather than treated as proof of compatibility.
PC Build Generation	High	The system can filter and rank suitable component combinations based on budget, intended use, and predefined rules.
System Performance	High for a Prototype	A basic web server and relational database should be sufficient for a classroom demonstration. Performance at a larger scale will require further testing.
Data Visualization	High	Charts and comparison tables can help users understand component costs, estimated performance, and differences between builds.
Data Accuracy	Moderate	Reliable recommendations depend on accurate component specifications, regularly reviewed data, and correctly implemented compatibility rules.

Technical conclusion: feasible, with data quality and rule validation as the main engineering challenges. An early proof of concept should validate the component schema, compatibility checks, and candidate generation before building extensive interface features.

5.2 Operational Feasibility

To make the application easy for beginners, the interface will use simple language, explain technical terms, and display clear warnings when compatibility problems are found. The warnings will also explain the problem and suggest what the user should check. Keeping prices and product availability up to date may become difficult because market information changes frequently. Therefore, the first version will display recorded prices with their sources and update dates rather than promise live market information.

The team will evaluate usability through supervisor feedback and a small walkthrough with potential users. This will help identify confusing features and improve the interface before the final demonstration. However, these observations will provide initial feedback rather than statistically conclusive evidence of user satisfaction.

Operational Conclusion: SmartPC Advisor is operationally feasible for a three-month academic project. Its success will depend on a simple interface, clear compatibility warnings, manageable data maintenance, and regular testing with potential users.

5.3 Economic and Financial Feasibility

SmartPC Advisor is economically feasible as a university project because the team can use free development tools, open-source software, and existing computers. However, some expenses may arise from hosting, domain registration, database services, backups, and project materials.

The following table presents an estimated budget for developing and running the prototype.

Cost Category	Description	Estimated Cost (BDT)
Domain and DNS	Registering a domain and configuring DNS for one year.	1,500–3,000
Hosting and Database	Hosting the application and database, assuming suitable student or free-tier services are available.	12,000–30,000/year
Backups, Monitoring, and Email	Basic services for data backup, system monitoring, and project communication.	4,000–12,000/year
Project Materials	Printing, additional internet usage, and presentation materials.	5,000–12,000
Contingency	Additional funds for unexpected expenses.	5,000–11,000
Estimated Direct Cash Cost	Total estimated one year project expenses.	27,500–68,000
Student Labor	Estimated value of development time contributed by the team. Opportunity cost of BDT 850–1,650 per hour.	300,000–600,000

Total Estimated Economic Cost : 327,500–668,000

Cost Analysis: The project can be developed at a relatively low cash cost because the team does not need to purchase expensive software or specialized hardware. Existing student computers and free development tools should be sufficient for the initial version. The estimated student labor represents the value of the time invested in planning, development, testing, and documentation. It is not a direct payment that the team must make.

If student email provides hosting, a domain, internet access, or other resources, the actual cash expenditure may be lower. However, paid component data, commercial services, or production-level hosting could increase the cost. The team should confirm actual prices before making purchases and prioritize free services wherever practical.

Economic Conclusion: SmartPC Advisor is economically feasible for an academic project, provided that the team controls its scope, uses available resources, and avoids unnecessary paid services. The proposed budget should be treated as an initial estimate and revised when actual requirements are known.

5.4 Schedule Feasibility

SmartPC Advisor is expected to be completed within 12 weeks (three months) as part of a semester project. With three team members, the project is achievable if tasks are divided properly, the team focuses on essential features, and progress is reviewed regularly.

The development process will be divided into six phases, from initial planning to final presentation. Each phase will have specific tasks and deliverables to help the team complete the project on time.

SI.	Phase	Timeline	Main Activities and Deliverables
1.	Planning and Requirements	Weeks 1–2	Define project objectives, identify core features, determine system requirements, and finalize the project scope.
2.	Database and System Design	Weeks 3–4	Design the database, collect initial component data, define compatibility rules, and prepare interface designs.
3.	Backend and Recommendation System	Weeks 5–6	Develop component management, filtering, compatibility checking, and the basic PC recommendation system.
4.	Frontend Development	Weeks 7–8	Build the user interface, display recommended PC builds, add comparison features, and create charts.
5.	Integration and Testing	Weeks 9–10	Connect the frontend and backend, test compatibility rules, identify errors, and improve system reliability.
6.	Evaluation and Final Submission	Weeks 11–12	Conduct a small usability walkthrough, complete documentation, prepare the final report, and demonstrate the application.

To keep the project on schedule, the three team members should divide responsibilities according to their skills while collaborating on integration and testing. The team should also review its progress at the end of Week 2 to confirm that the selected features, technologies, and workload are realistic. By the end of Week 6, the basic recommendation system should be working. If development falls behind, the team should reduce the component catalog or simplify comparison features instead of adding more time-consuming features.

Schedule Conclusion: SmartPC Advisor is moderately feasible within the 12-week semester. Successful completion will depend on clear task distribution, regular progress reviews, early testing, and prioritizing essential features over optional improvements.

5.5 Legal, Ethical and Data Feasibility

The planned system can minimize personal data by allowing anonymous use and avoiding accounts in the initial release. Product specifications and price observations should record their source and date. Public availability does not automatically grant reuse rights for images, text, or database compilations. Prefer manufacturer-published factual specifications, permitted feeds, original descriptions, or appropriately licensed material; attribute sources and review terms before reuse.

Because deployment may occur in Bangladesh, the project team should review applicable local requirements before collecting personal information or using overseas hosting. The official Bangladesh Laws portal lists the Personal Data Protection Act, 2026 (Act No. 63 of 2026), including provisions on classification, storage, and transfer; applicability, commencement, rules, and obligations should be

checked with the institution or qualified counsel before deployment. This study is not legal advice. Copyright and attribution practices should also be reviewed under the Copyright Act, 2023.

Ethical recommendations should explain criteria, avoid hidden retailer promotion, identify sponsored or affiliate relationships if ever introduced, and give equivalent treatment to products when data quality permits. Price and performance rankings can disadvantage products with incomplete data; missing values should be disclosed rather than silently treated as poor performance.

Conclusion: SmartPC Advisor can be designed to follow responsible data and ethical practices. The team should prioritize anonymous use, reliable data sources, proper attribution, transparent recommendations, and a review of applicable legal requirements before deployment.

5.6 Security and Privacy Considerations

SmartPC Advisor will be developed with basic security and privacy measures to protect the application and its data. Since the project is designed for a three-month semester and does not require user accounts in its first version, the amount of personal information collected will be minimal.

The team will follow these security practices:

1. **Minimal Data Collection:** The application will not require users to create accounts or provide personal information to generate PC builds.
2. **Secure Database Operations:** The backend will use parameterized database queries and validate user inputs to reduce the risk of SQL injection and other common attacks.
3. **Administrative Access Control:** If component management features are accessible through a hosted application, authentication and appropriate permissions will be required to prevent unauthorized changes.
4. **Secure Communication:** HTTPS will be used for hosted deployments to protect data transmitted between users and the application. Sensitive configuration values, such as passwords and API keys, will not be stored in the source code.
5. **Software Maintenance:** The team will update dependencies when necessary, restrict database permissions, handle errors safely, and maintain regular backups.
6. **Future Privacy Requirements:** If user accounts are introduced in future versions, the team will establish clear policies for data collection, consent, storage, deletion, access control, and incident response before implementation.

Security Conclusion: SmartPC Advisor can maintain relatively low privacy risks by allowing anonymous use and avoiding unnecessary personal data collection. However, even a small hosted application requires appropriate security controls. Our team will conduct basic security checks before the final demonstration. These checks will help identify common vulnerabilities, but they will not constitute a professional penetration test or guarantee that the application is completely secure.

6. Schedule and Resource Feasibility

6.1 People and Roles

The project will be developed by three students over 12 weeks. Each member will have specific responsibilities, with collaboration during integration and testing.

Team Member	Main Responsibilities
Member 1	Project planning, requirements, frontend development, and documentation.
Member 2	Backend development, database design, and API implementation.
Member 3	Component data collection, compatibility rules, recommendation logic, and testing.
All Members	Integration, debugging, evaluation, and final presentation.

In a small student team, one person may hold multiple roles. The 360–480 hours used for schedule planning are an approximate total team effort; role allocations overlap and should not be added to produce a larger total.

6.2 Infrastructure and Tools

The project will use the following resources:

1. **Hardware:** Existing student computers and internet access.
2. **Development Tools:** Git, GitHub, and a code editor.
3. **Technologies:** React or Next.js, FastAPI, and PostgreSQL or MySQL.
4. **Hosting:** A suitable free or low-cost hosting service for demonstration.
5. **Component Data:** Manufacturer specifications and dated price records.
6. **Testing:** Web browsers, test cases, and sample PC builds.

Conclusion: The project is achievable within 12 weeks if the three members divide responsibilities effectively, use available resources, and focus on essential features.

7. Legal, Ethical, Data and Security Considerations

7.1 Data Governance

SmartPC Advisor will manage component information, prices, and user inputs responsibly.

Data Type	Examples	How It Will Be Managed
Component Data	Socket, RAM type, dimensions, connectors, and power requirements.	Record sources, units, verification dates, and reviewer notes.
Price Data	Product prices and availability.	Record prices with their sources and dates; do not guarantee current availability.
User Inputs	Budget, intended use, and preferences.	Process inputs without saving personal information in the initial version.
System Records	Errors and administrative changes.	Limit access and avoid storing unnecessary personal information.

7.2 Ethical Use and Accessibility

The interface should present recommendation reasons, total cost, key trade-offs, and any unknown compatibility fields. It should provide keyboard access, meaningful labels, sufficient contrast, and chart alternatives that communicate the same information in text. The project should not claim conformance without an evaluation.

Conclusion: The project will prioritize responsible data management, transparent recommendations, user privacy, and accessibility. These practices will help make SmartPC Advisor more trustworthy and easier to use.

8. Risk Assessment

Risk	Impact	Mitigation Strategy
Incorrect or outdated component data	Medium - High	Verify important specifications, record sources and dates, and flag missing information.
Compatibility checking errors	Medium - High	Test compatibility rules and display warnings for uncertain cases.
Project delays or scope expansion	High - Medium	Finalize core features by Week 2 and prioritize essential functions.
Inaccurate or outdated prices	High - Medium	Display prices with their sources and dates instead of promising live prices.
Unfair or unclear recommendations	Medium - Medium	Explain scoring criteria, weights, and limitations.
Copyright or licensing issues	Medium - High	Check usage permissions and provide appropriate attribution.
Security problems or data loss	Medium - High	Restrict administrative access, use HTTPS, protect credentials, and maintain backups.
Team availability or delayed feedback	Low - Medium	Assign responsibilities, track progress, and schedule regular reviews.
Accessibility problems	Low - Medium	Test keyboard navigation, text labels, color contrast, and chart alternatives.
Users misunderstand estimates	Medium - High	Clearly label estimated prices, power requirements, and performance, and advise users to verify compatibility before purchasing.

Conclusion: The main risks are inaccurate component data, compatibility errors, and project delays. The team can reduce these risks through careful planning, regular testing, transparent recommendations, and a clearly defined project scope.

9. Constraints, Benefits and Limitations

9.1 Constraints

1. **Limited Time and Resources:** The team has three members and 12 weeks to complete the project.
2. **Data Availability:** Component specifications and prices may be incomplete or outdated.
3. **Limited Performance Data:** The project does not have a validated dataset for predicting real-world PC performance.
4. **Compatibility Limitations:** The system can check only the compatibility rules implemented using available component data.
5. **Budget and Infrastructure:** Hosting costs, available services, and university requirements may affect development.

9.2 Expected Benefits

1. **Easy PC Building:** Helps beginners select suitable components for their needs.
2. **Time Saving:** Reduces the need to research and compare every component manually.
3. **Compatibility Checking:** Identifies known compatibility issues before purchasing.
4. **Budget Management:** Shows the total build cost and how the budget is distributed.
5. **Better Comparisons:** Helps users compare different builds based on price, estimated performance, and value.
6. **Future Development:** Provides a foundation for adding more features and improving recommendations later.

9.3 Limitations

Recommendations can only be as accurate as the component records, rules, and price observations used. A “compatible” result means that the implemented checks passed; it does not verify physical assembly, BIOS version, firmware readiness, product revisions, local stock, warranty conditions, or all manufacturer-specific caveats. Price/performance scores are comparative indicators, not universal quality judgments. Estimated power is not a measurement. Market coverage, usability, recommendation quality, and production security remain unverified until evaluated.

Overall, SmartPC Advisor will provide practical guidance for PC building, but users should verify component specifications, prices, availability, and compatibility before purchasing.

10. Conclusion and Recommendation

SmartPC Advisor is feasible as a university prototype if its initial release remains a transparent, rule-based build planner with a bounded product catalog. The selected stack is widely used for web applications, the main feature set is implementable within a 12-week academic schedule, and expected direct costs are manageable under the stated assumptions. The largest uncertainties are source quality, correctness of compatibility logic, project staffing, and the freshness of local prices.

Recommendation Based on the feasibility report: proceed to requirements confirmation and a small proof of concept, conditional on (1) a week-two scope and data-source approval, (2) a documented

compatibility ruleset with explicit unknown handling, (3) a named data steward, (4) a week-six working end-to-end flow, and (5) a final review that describes results and limitations honestly. If these conditions cannot be met, reduce catalog size and defer features before increasing cost or claiming readiness.

This recommendation is for academic planning and demonstration. A public or commercial service would require a separate review of legal obligations, data rights, security, support operations, uptime, retailer relationships, and user-facing liability.